BAHIR DAR UNIVERSITY

BAHIR DAR INSTITUTE OF TECHNOLOGY(BIT)

FACULTY OF COMPUTING

Operating System And System Programming INDIVIDUAL PROJECT

TITLE: Installation of HarmonyOS

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1.Introduction

An operating system (OS) is essential system software that manages computer hardware and software resources and provides services for application programs. It acts as an intermediary between the user and the physical components of a computer, such as the processor, memory, and input/output devices. Through the operating system, users can run applications, manage files, control hardware devices, and perform various tasks efficiently. Without an operating system, a computer or smart device cannot function or execute user commands effectively.

Operating systems also play a critical role in resource management. They ensure that hardware resources are allocated properly, prevent conflicts between programs, and maintain system stability and security. Common examples of operating systems include Windows, Linux, macOS, and Android, each designed for specific platforms and purposes.

HarmonyOS is a modern and innovative operating system developed by Huawei. It is designed to operate across a wide range of devices, including smartphones, tablets, smartwatches, smart TVs, and Internet of Things (IoT) devices. Unlike traditional operating systems that are mainly focused on a single type of device, HarmonyOS is built with a distributed architecture. This allows multiple devices to connect, communicate, and share resources seamlessly, functioning together as a unified system.

One of the key features of HarmonyOS is its distributed technology, which enables users to control multiple devices as if they were one. For example, a user can start a task on a smartphone and continue it on a tablet or smart TV without interruption. This creates a smooth and integrated user experience across different platforms.

The importance of HarmonyOS lies in its ability to provide a flexible, efficient, and scalable platform for modern smart environments. It enhances device connectivity, improves system performance, and supports the development of cross-device applications. In addition, it offers strong security features and efficient resource management, making it suitable for the growing ecosystem of smart devices.

As technology continues to advance and the number of connected devices increases, operating systems like HarmonyOS play a crucial role in enabling better communication, integration, and user experience. It represents a step forward in the evolution of operating systems by focusing on interoperability, distributed computing, and future-ready smart solutions.

2. Objectives

The main objective of this project is to attempt the installation of HarmonyOS in a virtual environment using available tools and technologies. This project aims to provide a clear understanding of the fundamental concepts and functions of an operating system, including how it manages hardware and software resources.

Another important objective is to gain practical experience in setting up and working with development tools related to HarmonyOS, such as configuring the required environment and understanding the installation process. The project also focuses on learning how virtual environments are used to test and run operating systems without affecting the host system.

In addition, this project aims to study the architecture and key features of HarmonyOS, particularly its distributed system design, which allows multiple devices to work together seamlessly. It also seeks to compare HarmonyOS with other operating systems in terms of performance, flexibility, and usability.

Furthermore, the project is intended to identify and analyze the challenges faced during the installation process and to improve problem-solving skills by finding appropriate solutions to technical issues. It also aims to develop skills in system configuration, troubleshooting, and documentation by recording each step of the installation process clearly.

3. Requirements

3.1 Hardware Requirements

The following hardware specifications are required and were used for working with HarmonyOS:

- Laptop/Desktop computer
  This device is used for installing and running HarmonyOS development tools and applications.
- 4 GB RAM
  This is the minimum memory capacity used. It is enough for basic installation, but performance becomes slower when running the emulator or multiple applications.
- At least 10 GB free storage space
  Required for installing DevEco Studio, HarmonyOS SDK, and project files.

- Intel Core i3 processor (or equivalent)
  Used for processing installation and development tasks. It supports basic operations but may not handle heavy simulation smoothly.
- Stable power supply
  Ensures the system remains on during installation and prevents interruption or data loss.

3.2 Software Requirements

The software environment used is based on HarmonyOS as the operating system, along with development tools for application building:

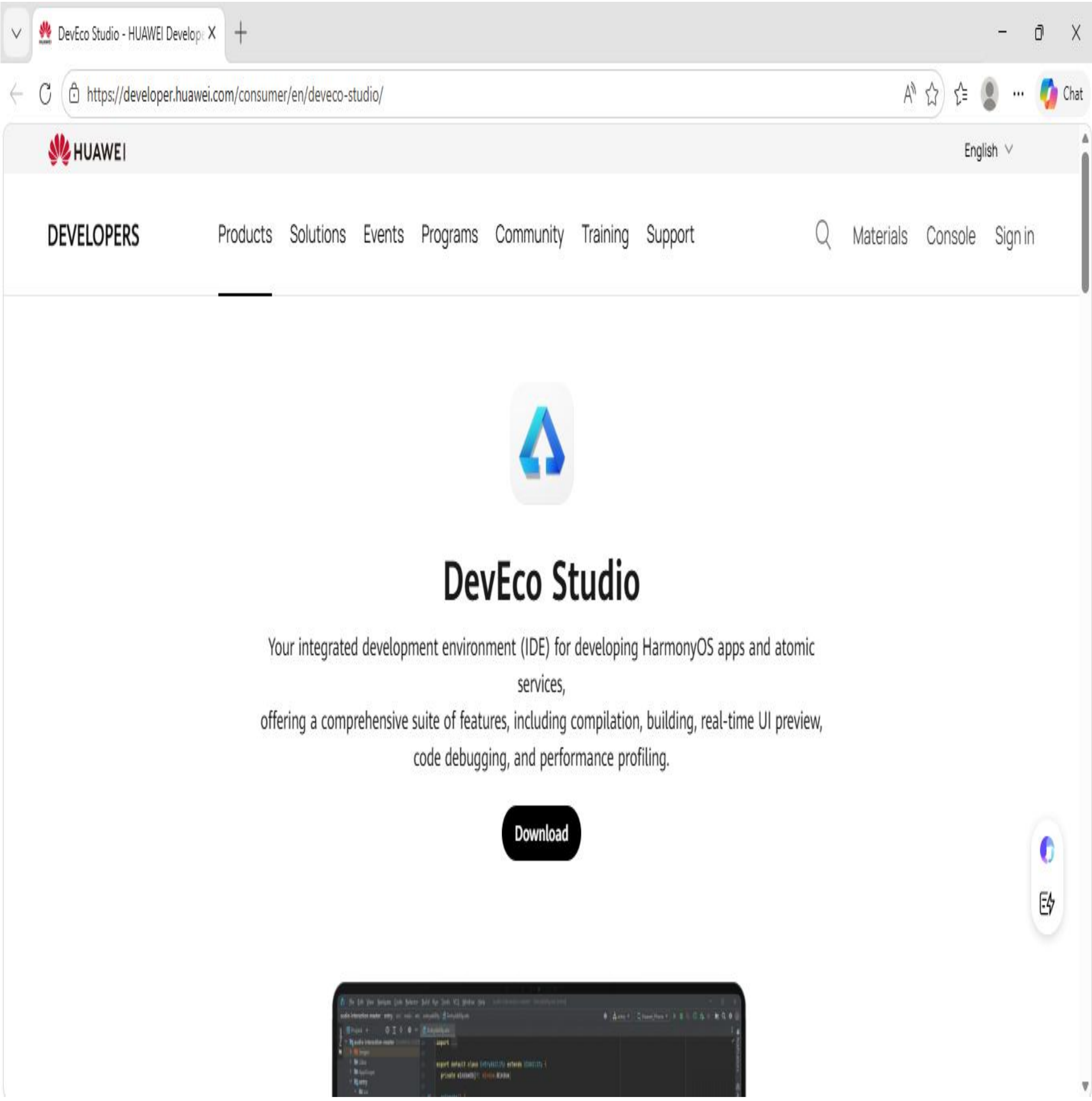
- HarmonyOS (Operating System)
  The main operating system used for development and testing. It provides the environment where applications and system tools run.
- DevEco Studio
  The official development tool used to create, build, and manage HarmonyOS applications.
- HarmonyOS SDK (Software Development Kit)
  A collection of tools, libraries, and APIs required to develop applications for HarmonyOS.
- Built-in Emulator (DevEco Studio)
  A virtual device used to simulate HarmonyOS devices for testing applications without needing a physical device.

4. Installation Steps (Attempted Installation)

The following steps describe the process followed to set up and attempt running HarmonyOS using DevEco Studio.

Step 1: Download DevEco Studio

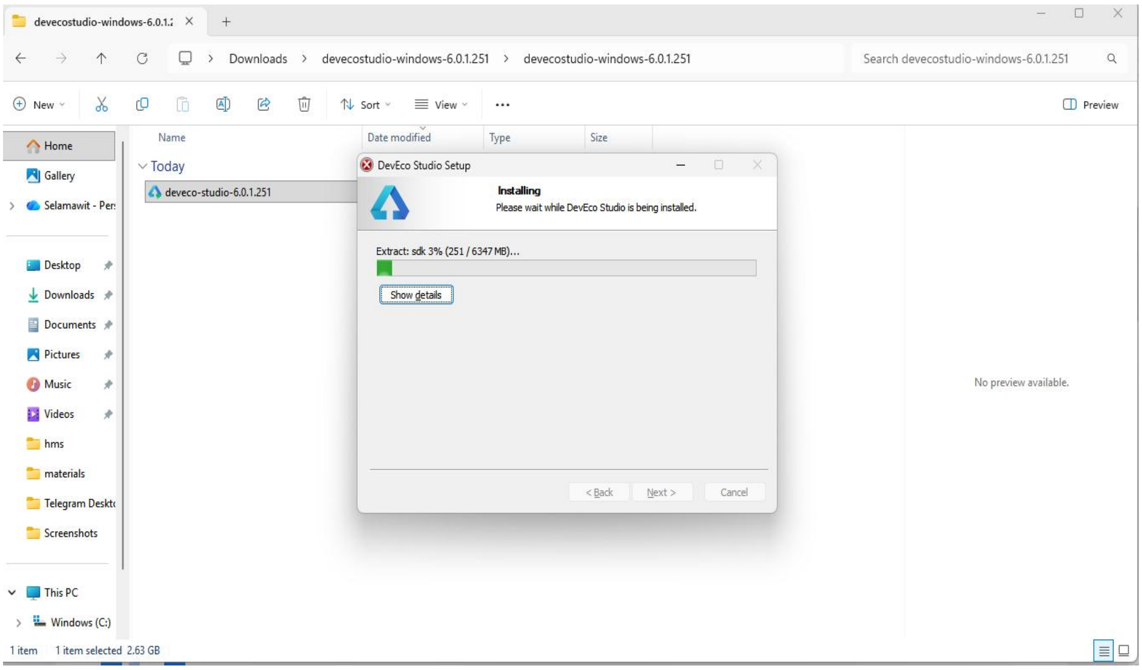
DevEco Studio was downloaded from the official Huawei developer website. The installer package was saved locally on the computer for installation.

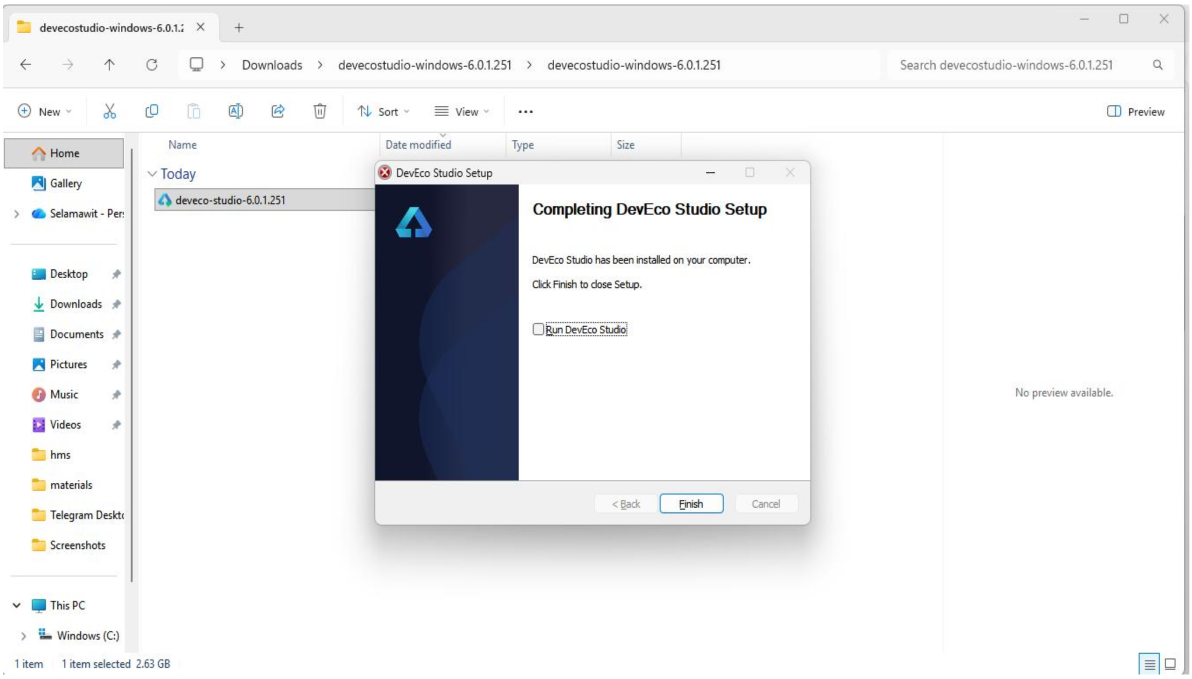


Step 2: Install DevEco Studio

The downloaded installer was opened and the installation process was followed step by step:

- Selected Next → Install → Finish during the setup wizard
- Default installation settings were used
- The installation process was completed successfully

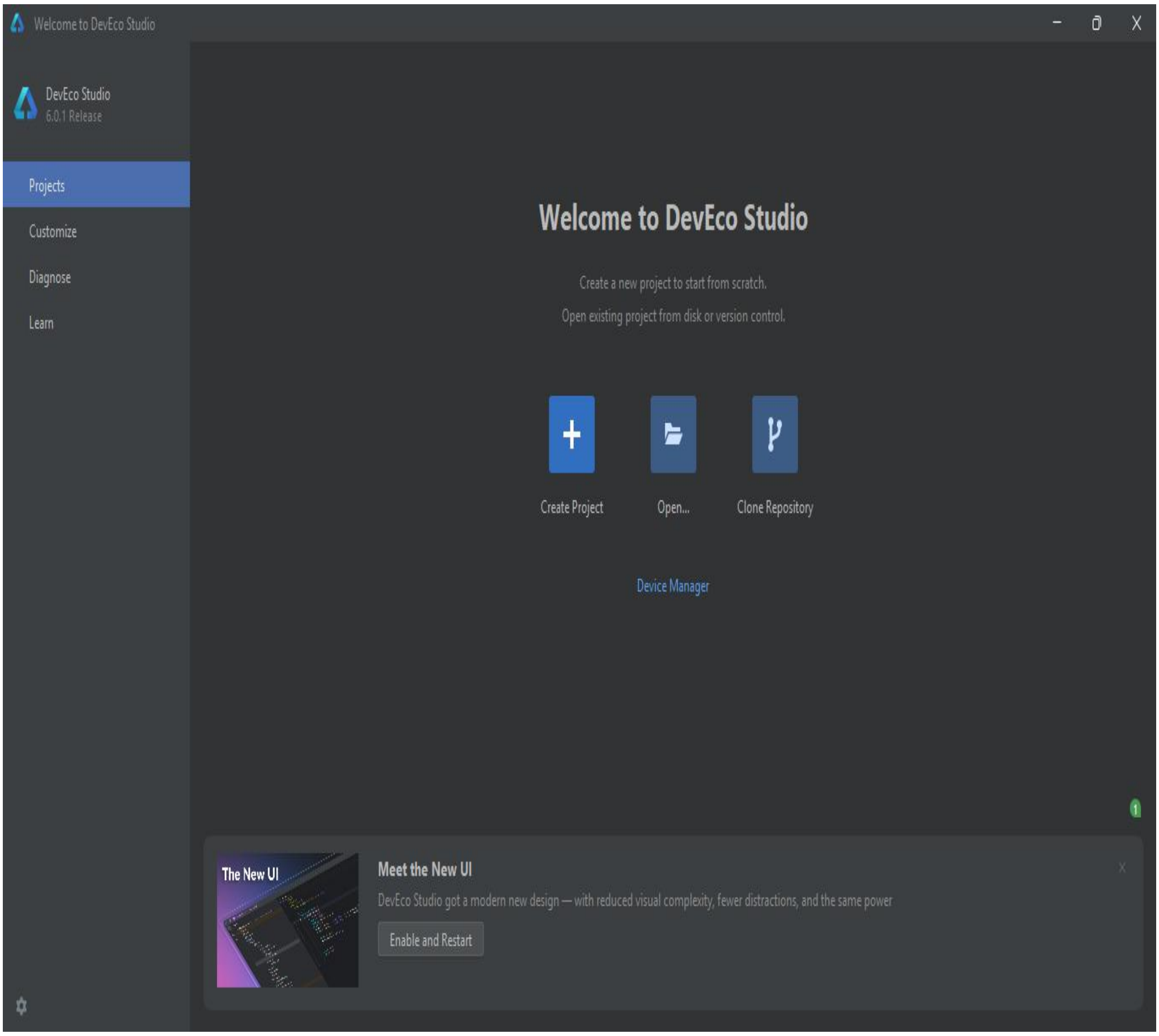




Step 3: Launch DevEco Studio

DevEco Studio was opened after installation.

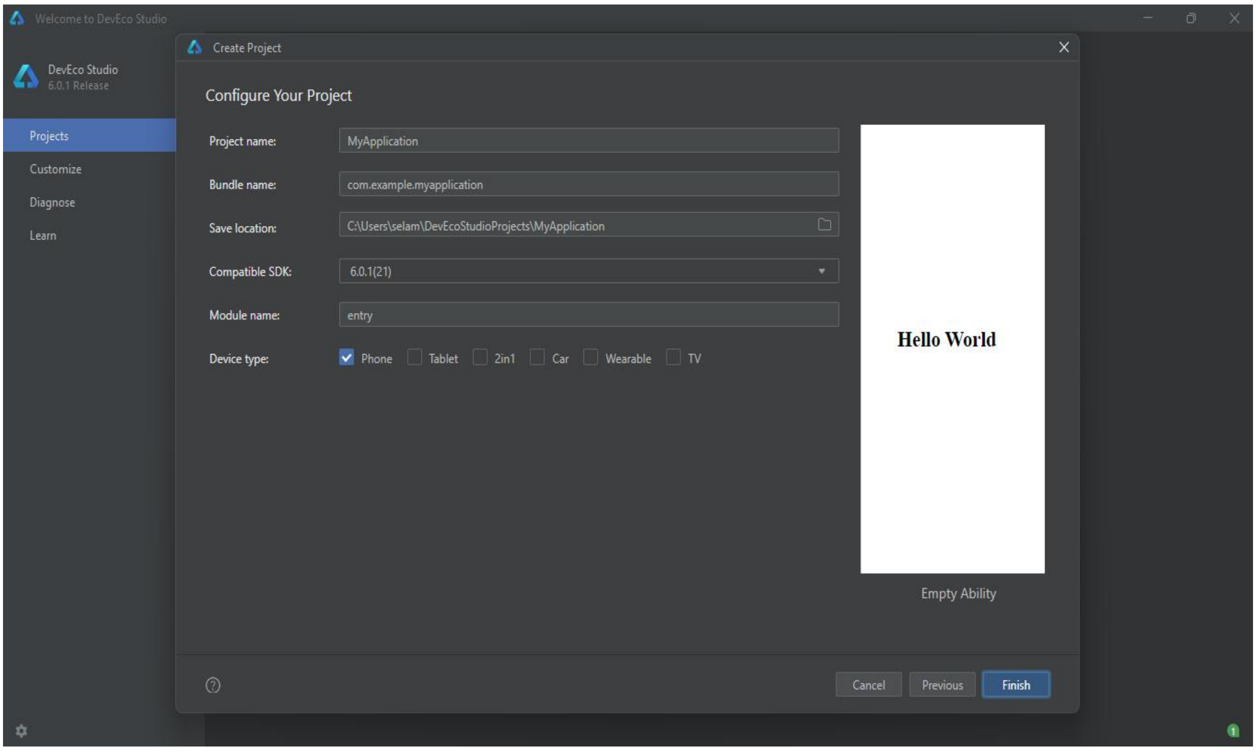
- Initial configuration process started automatically
- Required system components and dependencies were downloaded
- Setup completed successfully before first use



Step 4: Install HarmonyOS SDK

The SDK Manager in DevEco Studio was opened to install the required development tools.

- The HarmonyOS SDK packages were selected from the available list
- The download process was started and completed through the SDK Manager
- After downloading, the packages were automatically installed into the system
- These SDK components provide the necessary libraries and tools for developing and building HarmonyOS applications


Welcome to DevEco Studio
DevEco Studio
6.0.1 Release
Projects
Customize
Diagnose
Learn
Create Project
Configure Your Project
Project name:
MyApplication
Bundle name:
com.example.myapplication
Save location:
C:\Users\selam\DevEcoStudioProjects\MyApplication
Compatible SDK:
6.0.1(21)
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entry
Device type:
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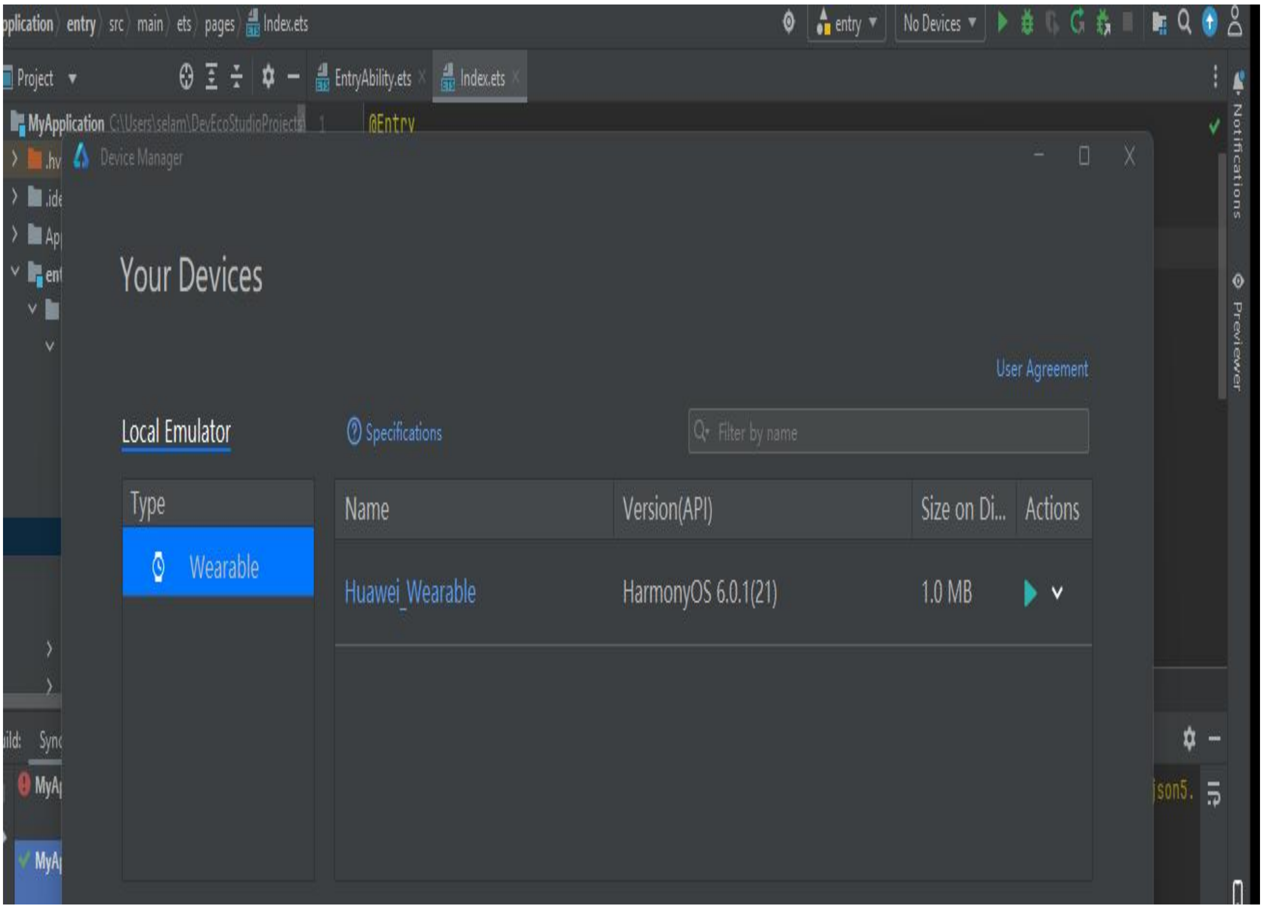
entry
src
main
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Index.ets
No Devices
Project
EntryAbility.ets
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MyApplication
@Entry
Device Manager
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Specifications
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Type
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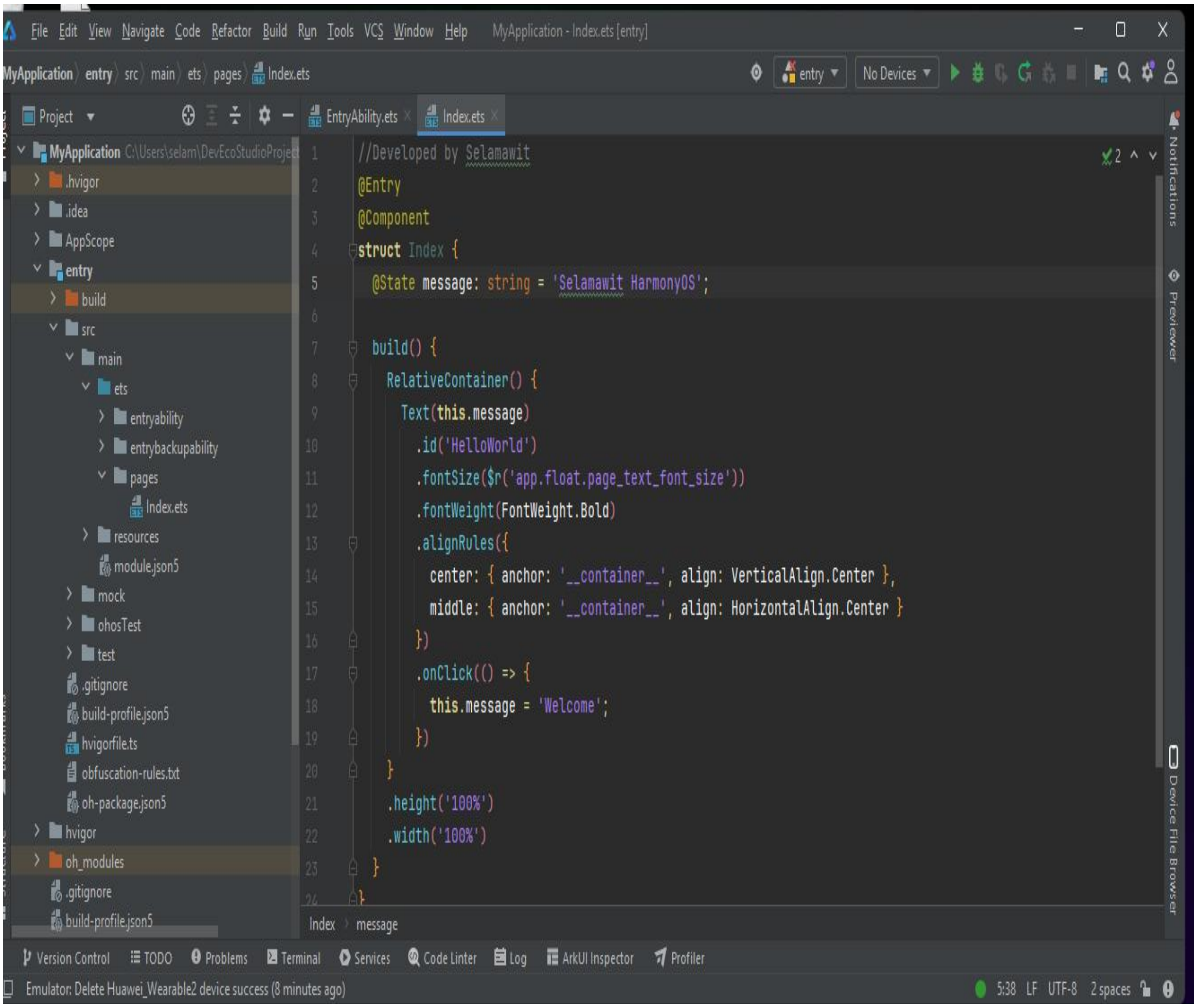
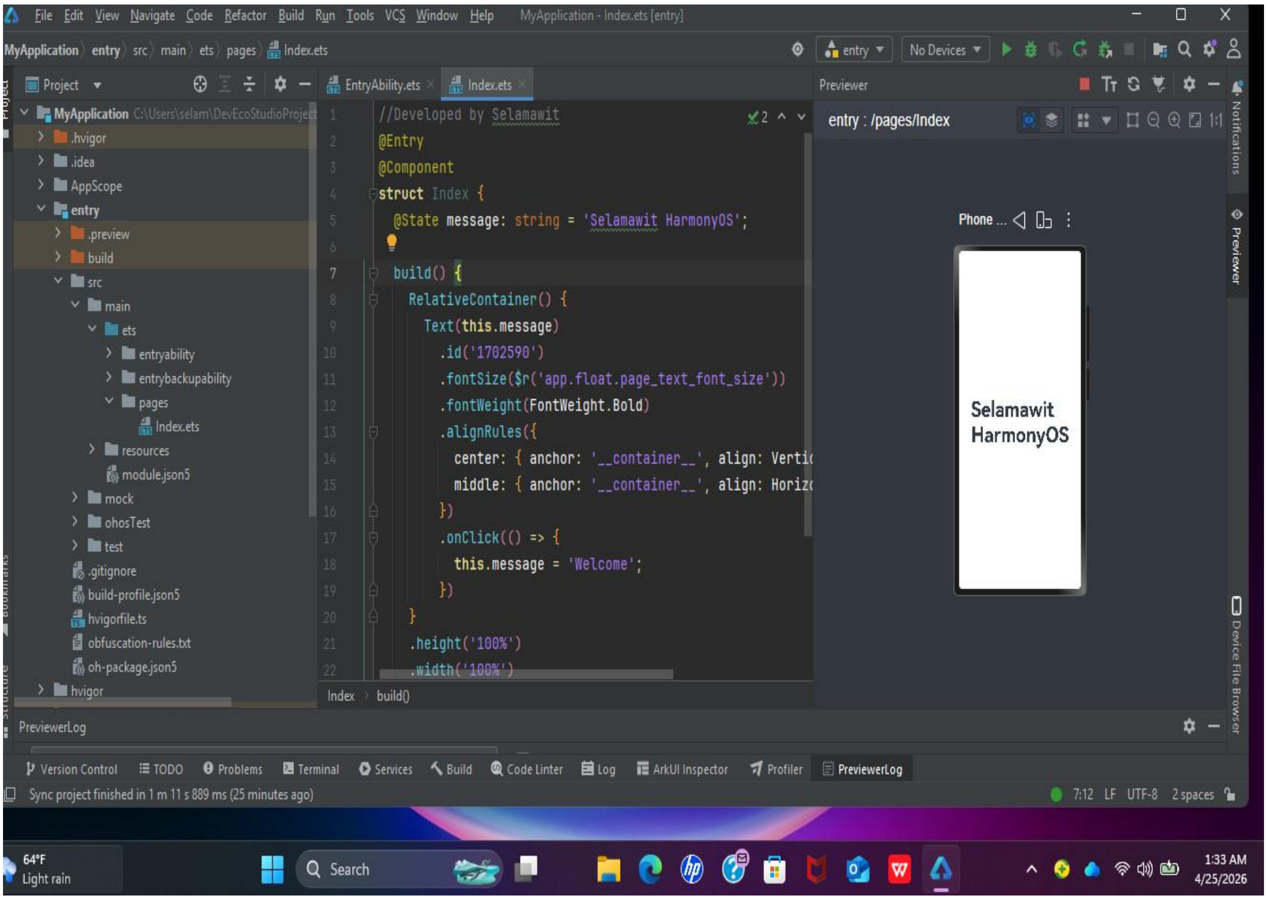
Name
Version(API)
Size on Di...
Actions
Huawei_Wearable
HarmonyOS 6.0.1(21)
1.0 MB
Notifications
Previewer

Step 5: Create a New Project

A new project was created in DevEco Studio to begin application development.

- The "Create Project" option was selected from the start menu
- A suitable HarmonyOS application template was chosen
- Project details were entered, including the project name and required basic information
- The project structure was automatically generated by the system based on the selected template





Step 6: Set Up Emulator

The Device Manager in DevEco Studio was used to configure a virtual testing environment.

- The Device Manager was opened from DevEco Studio
- A new virtual device (emulator) was created
- Required settings such as RAM and storage allocation were configured
- The emulator was prepared to simulate a HarmonyOS device for application testing

```text
- CPU: x86_64 architecture (Intel/AMD)
- RAM: Minimum 8GB (16GB recommended)
- Disk: At least 50GB free space
- Virtualization: Enabled in BIOS (VT-x/AMD-V)
```

Step 7: Run Application (Attempt)

The project was executed to test the configured environment.

- The Run (   ) button in DevEco Studio was selected
- The emulator was triggered to start
- The system attempted to launch the HarmonyOS runtime environment
- Application execution process began for testing purposes

Step 8: Installation Result

The final outcome of the installation attempt is summarized below:

- The installation process was not completed successfully
- The HarmonyOS environment did not run on the PC system
- The emulator failed to fully launch the operating system

Reason:
This limitation is mainly due to compatibility and system requirements. HarmonyOS development and execution are optimized for supported devices and specific virtual environments. Standard PC configurations may not fully support its runtime environment.

5. Issues (Problems Faced)

During my attempt to install and run HarmonyOS, I encountered several problems step by step:

First, after completing the installation of DevEco Studio and setting up the environment, I tried to proceed with running HarmonyOS. However, the system did not fully install or launch on my PC.

Next, I realized that HarmonyOS does not have direct support for installation on standard personal computers like mine. This made it difficult to continue beyond the setup stage.

As I moved forward, I faced challenges in running the operating system outside of officially supported devices or environments. The system requires specific configurations or hardware support that my PC could not provide.

Finally, when I set up the emulator and attempted to run it, the emulator did not successfully start a complete HarmonyOS system. It failed to fully load or function as expected.

6. Solutions

To handle the problems I faced, I took the following steps in sequence:

First, I successfully installed and set up DevEco Studio as the main development environment. This allowed me to at least access the tools required for HarmonyOS development.

Next, I completed the initial configuration and explored the features available within DevEco Studio to better understand how the environment works.

After that, I attempted to continue with the HarmonyOS setup process. However, due to the lack of full support for PC environments, I was unable to proceed to full installation or execution.

Because of this limitation, I decided to stop at the development environment stage instead of forcing the installation.

Finally, I focused on understanding and documenting the entire setup process, including the steps I followed and the limitations I encountered. This helped me complete the assignment even though the system could not fully run.

7. Filesystem Support in HarmonyOS

In HarmonyOS, different filesystems are used to help the operating system store, organize, and manage data properly. A filesystem is basically a method that controls how files are saved, read, and accessed on storage devices like internal memory, SD cards, or USB drives.

HarmonyOS uses more than one filesystem because each one is designed for a specific purpose.

7.1 ext4 (Extended File System 4)

The ext4 filesystem is mainly used for the internal storage of HarmonyOS, especially for system files and installed applications.

It is widely used in Linux-based systems, and since HarmonyOS is also built on similar technology, ext4 fits very well.

Why HarmonyOS uses ext4:

- It is very stable and reliable
- It supports large files and large storage systems
- It helps reduce data corruption and keeps the system safe
- It provides good performance for system operations

7.2 FAT32 (File Allocation Table 32)

FAT32 is commonly used for external storage devices such as USB flash drives and SD cards. It is also used when transferring files between different devices.

Why HarmonyOS uses FAT32:

- It is compatible with almost all operating systems (Windows, Linux, Android, etc.)
- It makes file sharing very easy between devices
- It is simple and widely supported

7.3 F2FS (Flash-Friendly File System)

F2FS is designed specifically for flash memory storage, which is commonly used in smartphones and modern devices.

Why HarmonyOS uses F2FS:

- It is optimized for flash storage (like phone memory)
- It improves reading and writing speed
- It reduces wear on storage chips, helping them last longer
- It enhances overall device performance

To summarize, HarmonyOS uses different filesystems based on the type of storage and its purpose:

- ext4 → used for internal system storage because it is stable and reliable
- FAT32 → used for external storage because it is highly compatible
- F2FS → used for flash storage because it improves speed and durability

Overall, HarmonyOS chooses these filesystems to ensure a balance between performance, compatibility, and system reliability, depending on where and how the data is stored.

8. Advantages & Disadvantages

HarmonyOS is a modern operating system that is designed to work on different smart devices like phones, tablets, TVs, and other smart devices. It focuses more on connection between devices and smooth performance. But like other systems, it also has both advantages and disadvantages.

8.1Advantages of HarmonyOS

- Fast and smooth performance
  HarmonyOS is designed to run smoothly and respond quickly. It manages system resources in a good way, so even devices with average specifications can still work without serious lag.
- Works on many devices (one ecosystem idea)
  One of the main advantages is that it is not limited to just smartphones. It can work on different devices like smart TVs, tablets, watches, and IoT devices. This makes it easy for all devices to be connected together in one system.
- Easy connection between devices
  It allows devices to communicate with each other easily. For example, files or data can be shared between devices without much difficulty, and tasks can continue from one device to another.
- Better security and stability
  HarmonyOS is built with strong security features. It helps to protect user data and keeps the system stable, reducing chances of crashes or errors.
- Suitable for smart technology
  It is designed for modern smart devices, so it works well in environments where many devices are connected together.

8.2Disadvantages of HarmonyOS

- Limited applications
  Compared to Android or Windows, HarmonyOS still has fewer apps available. Some popular applications are not fully supported or are missing, which can be a problem for users.
- Difficult installation process
  In real situations, HarmonyOS is not something that can be easily installed like normal software. It depends on device compatibility and official support, so it cannot always be installed freely on any computer or device.

- Compatibility issues
  Some apps or tools made for other operating systems may not work properly on HarmonyOS.
- Still developing system
  Since HarmonyOS is still new, it is not fully developed yet. Some features are still improving, and it is not as widely used as other operating systems.

9. Conclusion

HarmonyOS is a modern operating system designed to operate across a wide range of smart devices such as smartphones, tablets, smart TVs, and Internet of Things (IoT) devices. It focuses on delivering smooth performance, efficient resource management, and strong connectivity between devices within a unified ecosystem. Its distributed architecture allows multiple devices to work together seamlessly, improving the overall user experience.

Through this project, it was understood that HarmonyOS utilizes different file systems such as ext4, FAT32, and F2FS depending on the type of storage and device requirements. Each file system is selected based on factors like stability, compatibility, and performance optimization, which are essential for ensuring efficient system operations.

The project also provided insight into the practical aspects of installing HarmonyOS in a virtual environment. It highlighted the importance of meeting system requirements, properly configuring development tools, and following installation procedures carefully. During the process, several challenges were encountered, including compatibility issues, configuration errors, and limitations of running the system on unsupported platforms. These challenges helped in developing problem-solving and troubleshooting skills.

Furthermore, HarmonyOS offers several advantages, including fast performance, efficient multitasking, enhanced device interaction, and strong security features. Its ability to support cross-device collaboration makes it a promising solution for future smart environments. However, it also has some limitations, such as a relatively smaller application ecosystem, limited support on non-Huawei devices, and difficulties in installation on general-purpose computers.

In conclusion, this project successfully achieved its objectives by providing both theoretical knowledge and practical experience related to operating systems and HarmonyOS installation. It also improved technical understanding, documentation skills, and the ability to work with modern computing technologies. Overall, HarmonyOS represents a significant advancement in operating system design, especially in the area of distributed computing and smart device integration.

10. Future Recommendation

HarmonyOS is still a developing operating system, and it has a lot of potential for the future as technology continues to advance. With its distributed system design and focus on smart device connectivity, it can grow into a stronger platform if it keeps improving and gaining support.

- **It may grow and become more popular in the future**
  As more companies and device manufacturers adopt HarmonyOS, it can expand its ecosystem and become a stronger competitor to operating systems like Android, Windows, and iOS. If it is widely used, it can improve compatibility between different smart devices and create a more connected digital environment.
- **Needs more developer support**
  One of the main improvements needed for HarmonyOS is increasing developer participation. More developers should be encouraged to create applications for the platform. This will help solve the problem of limited applications and make the system more useful and attractive for users. A strong app ecosystem is very important for any operating system to succeed.

- **Should improve compatibility**
  In the future, HarmonyOS should support a wider range of third-party applications and tools. Better compatibility will allow users to run more software without restrictions. This will also help users switch from other operating systems more easily without facing application limitations.
- **Better installation and accessibility in the future**
  The installation process should become simpler and more user-friendly. If HarmonyOS can be easily installed on different devices, including PCs and non-Huawei hardware, more users will be able to access and try it. Easier accessibility will increase its popularity and usage.
- **Continuous improvement in features**
  HarmonyOS should continue to improve important features such as speed, performance, security, and multi-device connectivity. Regular updates and optimizations will help it stay competitive in the modern technology world. Enhancing system stability and reducing bugs will also improve user experience.
- **Expansion in IoT and smart ecosystems**
  Since HarmonyOS is designed for smart devices, it should focus more on Internet of Things (IoT) development. Better integration with smart homes, wearables, and industrial devices will make it more powerful and widely used in future technologies.
- **Stronger global adoption**
  For long-term success, HarmonyOS should expand beyond local markets and gain global acceptance. This requires better international support, language compatibility, and partnerships with global technology companies.

11. What is Virtualization?

Virtualization is a technology that allows one operating system (OS) to run inside another operating system using specialized software. Instead of installing a new OS directly on a computer's physical hardware, a virtual environment is created where the new system can run safely and independently.

In simple terms, virtualization means creating a "computer inside a computer." This virtual computer is called a **virtual machine (VM)**, and it behaves like a real physical system with its own virtual CPU, memory, storage, and network interface.

For example, HarmonyOS can be run on a laptop through an emulator or virtual machine, even if the laptop is originally running another operating system such as Windows. The virtual system looks and operates like a real device, but it is actually running within software on the host machine.

Virtualization works through a layer of software known as a **hypervisor**. The hypervisor is responsible for creating, managing, and running virtual machines. It controls how hardware resources such as CPU, RAM, and storage are shared between the host system and the virtual systems. There are two main types of hypervisors:

- **Type 1 (bare-metal):** Runs directly on hardware (e.g., VMware ESXi)
- **Type 2 (hosted):** Runs on top of an existing operating system (e.g., VirtualBox, VMware Workstation)

One of the key advantages of virtualization is **isolation**. Each virtual machine operates independently, so problems such as crashes or malware inside the virtual system do not affect the main (host) operating system. This makes virtualization very useful for testing, development, and learning purposes.

Additionally, virtualization improves **resource utilization** by allowing multiple operating systems to run on a single physical machine. It also provides flexibility, as users can easily create, delete, or modify virtual machines without affecting the actual hardware.

However, virtualization also has some limitations. Virtual machines may run slower than systems installed directly on hardware due to resource sharing. It also requires sufficient RAM, CPU power, and storage to function effectively.

11.1 Why Virtualization

Virtualization is widely used in computing because it makes working with different operating systems easier, safer, and more flexible.

One main reason is **safe testing**. Virtualization allows users to try new operating systems without affecting the real system. If something goes wrong, like a crash or error, it only affects the virtual machine, not the main computer. This makes it very safe for experiments and learning.

Another reason is that it **does not require extra physical computers**. Instead of using many computers for different operating systems, one computer can run multiple systems at the same time using virtual machines. This helps save money, space, and hardware resources.

Virtualization is also **easy to manage and control**. Virtual machines can be started, paused, restarted, or deleted at any time. It is also possible to save the current state and return to it later, which is very helpful during testing.

It is especially **useful for students and developers**. Students can use virtualization to practice and understand how operating systems work without risk. Developers can test their applications on different systems without needing many real devices.

Another benefit is **better use of resources**. A single computer can share its CPU, memory, and storage with multiple virtual machines, making the system more efficient.

Virtualization also provides **isolation and security**. Each virtual machine works independently, so problems like viruses or system errors in one machine do not affect others or the main system

11.2 How Virtualization Works

Virtualization works using special software called a **virtual machine (VM)** or an emulator. Common examples include VMware and VirtualBox. These tools allow a computer to run another operating system inside it without installing it directly on the physical hardware.

The main idea of virtualization is to divide the resources of a real computer such as CPU, RAM, storage, and network, and use them to create a **virtual computer** inside the system. This virtual computer behaves like a real system, even though it is running in software.

The real computer is called the **host system** (for example, a Windows laptop or desktop). The operating system running inside the virtual machine is called the **guest system** (for example, HarmonyOS, Linux, or Android). The virtualization software works as a bridge between the host and guest systems and manages how resources are shared.

Step-by-step explanation:

1. First, virtualization software like VMware or VirtualBox is installed on the main computer.
2. After installation, a new virtual machine (VM) is created inside the software.
3. During setup, the user assigns system resources such as CPU, RAM, and storage to the virtual machine.
4. The software then creates virtual hardware including:
    1. Virtual CPU
    2. Virtual RAM
    3. Virtual storage (hard disk)
    4. Virtual network adapter
5. An operating system such as HarmonyOS is installed inside the virtual machine using an installation file (ISO).
6. When the virtual machine is started, the operating system boots and runs like a normal computer system.

Although it looks like a separate computer, it is still using the real computer's hardware in a controlled way.

The virtualization software uses a component called a **hypervisor**, which is responsible for managing all virtual machines and distributing resources properly. It ensures that each virtual machine works independently without affecting others.

Multiple virtual machines can run at the same time on one physical computer. For example, a user can run Windows as the main system while also running HarmonyOS and Linux inside virtual machines.

Virtual machines can also be paused, restarted, or reset at any time. This makes it easy to test different operating systems and fix problems without affecting the main system.

Example:

If you want to run HarmonyOS:

- Your laptop is running Windows (host system).
- You install VMware or VirtualBox.
- You create a virtual machine inside it.
- You allocate memory and storage to it.
- You install HarmonyOS inside the virtual machine.
- Now HarmonyOS runs inside your laptop like a real device.

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